

deepest edge persists if
 $\nu \theta^{d-1} / (n + 1) > \rho_{\Sigma} / R_{\Sigma}$

$n = 10, \rho/R = 0.01$
 $1/11 = 0.0909$
single edge passes

$n = 100, \rho/R = 0.01$
 $1/101 = 0.0099$
single edge fails

depth is only meaningful
after the first edge clears
the persistence threshold

therefore table entry
 $d^* = 5$ for $n = 100, \rho/R = 0.01$
conflicts with the inequality

audit reading: the depth formula is useful, but the numerical illustration needs a boundary check